

Yili Wen

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RESEARCH INTERESTS

I design and build **wearable and hardware systems that restore and augment human movement**, working across **actuation and control, embedded sensing, and digital fabrication**. I prototype real hardware and evaluate it with the people it is built for. I want to extend this toward **assistive and rehabilitation robotics and exoskeletons**, and toward closing the loop between neural intent and adaptive physical assistance.

EDUCATION

- 2025 – Present **Graduate Student, Computer Science — The University of Texas at Dallas**
Design & Engineering for Making (DE4M) Lab, advised by Liang He.
M.S. in Computer Science (Interactive Computing) expected **Spring 2027**.
- 2021 – 2025 **B.A. in Computation and Design — Duke Kunshan University**
Dual-degree program with **B.A. in Interdisciplinary Studies, Duke University**.
Undergraduate research advisor: Xin Tong.
- Prior research positions:** Research Assistant with Xin Tong (HCI Lab, Duke Kunshan University & Information Hub, HKUST Guangzhou, 2023–2025) and Karen Cochrane (Embodied Computing Lab, University of Waterloo, 2023–2025).

PUBLICATIONS

- 2026 [E.1] ***Yili Wen**, Yafei Ge, Xin Tong, Liang He. “PufFab: Making Shape-Transformable Rice Paper for Playful Food Fabrication.” (Work-in-Progress; presented). *ACM TEI 2026*.
- 2026 [W.2] ***Yili Wen**, Ceci Verbaarschot, Liang He. “From Intention to Action: An Invasive BCI-Driven Wearable Framework for Everyday Hand Assistance in SCI Patients.” (Workshop paper & poster; presented). *Everyday Wearable for Personalized Health and Well-Being Workshop, ACM CHI 2026, Barcelona*.
- 2026 [P.1] Georgia Loewen, Karen Anne Cochrane, **Yili Wen**, Audrey Girouard. “Personas at Play: Understanding Disabled Player Experience to Expand Accessible Controller Design.” (Poster). *Graphics Interface 2026*.
- 2025 [C.1] ***Yili Wen**, Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. “‘I Want to Use Technology to Help Farmers in the Future’: Enhancing Children’s Understanding of Farming through Tangible User Interfaces and Embodied Interaction.” (Presented). *ACM Interaction Design and Children (IDC) 2025*.
- 2024 [C.2] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. “‘I Keep Sweet Cats in Real Life, But What I Need in the Virtual World Is a Neurotic Dragon’: Virtual Pet Design with Personality Patterns.” *Chinese CHI 2024*.
- 2023 [W.1] Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. “PetGen: Design and Generation of Virtual Pets.” (Workshop). *IEEE ICME Workshops (ICMEW) 2023*.

Manuscripts In Preparation

- ***Yili Wen**, Ceci Verbaarschot, Liang He. “A Soft Grasp-Assist Exoskeleton for Spinal Cord Injury: Design and Intent-Driven Control.” In preparation; targeting *IEEE Robotics and Automation Letters (RA-L)*.
- ***Yili Wen**, Liang He. “PACER: Cable-Driven Guidance for Independent Running by Blind and Low-Vision Runners.” In preparation; targeting *ACM IMWUT*.

SELECTED RESEARCH PROJECTS

- 2025 – Present **Soft Grasp-Assist Exoskeleton for Spinal Cord Injury** (“From Intention to Action”)
A wearable soft exoskeleton that restores grasping to people with spinal cord injury.
- Cast **three-chamber silicone finger actuators** from resin-printed molds, with a layer-jamming wrist brace for compliant grasp assistance during everyday tasks.
 - Built the **ESP32** pneumatic control stack that drives the three chambers independently, and integrated simulated neural-intent input from a collaborating invasive-BCI group.
 - Grounding the design in real rehabilitation needs through a clinician co-design study.

2025 – Present	PACER: Cable-Driven Wearable for Independent Blind & Low-Vision Running Blind and low-vision runners depend on a sighted volunteer guide holding a tether, so what limits their running is guide availability, not fitness. PACER is a body-worn cable-driven system that delivers the same directional pull runners already trust, so a run no longer needs a second person. • Built the cable actuation modules and ESP32 control stack that render direction as a pull, adapting the guide-rope tension cues runners already read, plus a custom object detector deployed on-device via Core ML in a SwiftUI companion app. • Ran two formative studies that set the cue vocabulary, five interviews with a blind running community in Chongqing, China, and a two-population survey (5 guide runners, 18 sighted runners) across a 19-situation taxonomy, which found the groups converge on <i>what</i> to do but diverge sharply on <i>how</i> it is conveyed.
2025	PuffFab: Shape-Transformable Rice Paper for Playful Food Fabrication (TEI '26) Modified a 3D printer and built the design tool for an edible-fabrication pipeline where moistening and cutting control how each region of a rice-paper sheet puffs into a raised 2.5D form when fried.
2024 – 2025	Adaptive Game-Controller Hardware for Upper-Limb Motor Disabilities (GI '26) Designed and fabricated the modular, remappable DIY controller prototypes that served as the physical platform for a study of disabled player experience.
2023 – 2024	WooGu: Tangible System for Children's Agricultural Literacy (IDC '25) Built a tangible farm-to-table learning game pairing sensor-embedded 3D-printed farming tools with an augmented display; evaluated with 15 children aged 5 to 12, improving their farming knowledge, engagement, and attitudes toward agriculture.

TEACHING & OUTREACH

Summer 2026	Instructor — TAST STEM Bridge & CS Outreach Summer Camps Department of Computer Science, UT Dallas. Six-week program (June–July 2026); designed and delivered hands-on lectures and build sessions.
2025 – 2026	Teaching Assistant — Department of Computer Science, UT Dallas CS 6326 Human–Computer Interaction (Fall 2025); CS 4376 Object-Oriented Design (Spring 2026).

TECHNICAL SKILLS

- **Fabrication & Prototyping:** soft-robotics fabrication (silicone molding/casting), SLA (resin) & FDM 3D printing, OpenSCAD / CAD, Blender, Figma.
- **Electronics & Embedded:** ESP32 / Arduino, soft pneumatic and DC-motor actuation, wireless (ESP-NOW).
- **Programming, ML & Vision:** Python, C/C++ (embedded), Swift/SwiftUI, JavaScript/p5.js, Java/Processing, R; custom object detection (YOLO / Ultralytics), on-device deployment (Core ML), OpenCV.
- **Research Methods & Analysis:** human-subjects studies (interviews, surveys, in-field evaluation), mixed methods, sensor-data analysis, statistics in R.

AWARDS

Center for the Study of Contemporary China (CSCC) Fellowship, Duke Kunshan University, 2023.
Dean's List, Duke Kunshan University — Fall 2022, Fall 2023, Spring 2024, Fall 2024.